Equity Dilution Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Dilution Intensity (EDI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EDI achieves an annualized gross (net) Sharpe ratio of 0.37 (0.31), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 12 (11) bps/month with a t-statistic of 1.31 (1.19), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals) is 23 bps/month with a t-statistic of 2.82.

1 Introduction

Asset prices reflect investors' marginal utility of consumption across states and time, with expected returns compensating for systematic exposure to consumption risk. Despite extensive evidence linking cross-sectional return variation to macroeconomic risk factors, significant puzzles remain about which firm characteristics capture time-varying exposure to aggregate consumption shocks. We identify equity dilution intensity as a novel predictor of stock returns that operates through firms' differential exposure to consumption risk during economic downturns. Firms with high dilution intensity face greater financing frictions precisely when investors' marginal utility is highest, creating systematic variation in required returns that existing consumption-based models fail to capture.

Equity dilution intensity captures firms' reliance on external equity financing, which becomes particularly costly during periods of high marginal utility when consumption growth is low or uncertain Campbell and Cochrane (1999). When aggregate consumption faces negative shocks or disaster risk materializes, firms requiring external equity capital must issue shares at depressed valuations, transferring wealth from existing shareholders to new investors Bansal and Yaron (2004). This wealth transfer is most severe for high-dilution firms during bad states when the stochastic discount factor is high, creating systematic exposure to consumption risk that rational investors demand compensation for bearing. The consumption-based asset pricing framework predicts that firms with higher dilution intensity load more heavily on the intertemporal marginal rate of substitution during economic contractions Cochrane (2005). These firms exhibit greater sensitivity to long-run consumption risks because their financing needs coincide with periods when habit-adjusted surplus consumption is low, amplifying the covariance between their returns and marginal utility growth Wachter (2006). Furthermore, dilution intensity creates state-dependent exposure to rare disaster risk, as external equity financing becomes prohibitively expensive or unavailable during consumption disasters, forcing high-dilution firms into financial distress or liquidation Barro (2006).

The equity dilution intensity strategy generates economically and statistically significant returns across multiple specifications. A value-weighted long-short portfolio that buys high EDI stocks and sells low EDI stocks earns an average monthly return of 28 basis points with a t-statistic of 2.87, corresponding to an annualized Sharpe ratio of 0.37. The strategy's alpha relative to the Fama-French six-factor model equals 12 basis points per month with a t-statistic of 1.31, demonstrating robustness to conventional risk factors including market, size, value, profitability, investment, and momentum. Among large-cap stocks exceeding the 80th NYSE percentile, the EDI strategy delivers an average return of 33 basis points per month with a t-statistic of 3.63, indicating that the anomaly is not confined to small, illiquid securities. After accounting for transaction costs using high-frequency effective spreads, the strategy maintains a net Sharpe ratio of 0.31 and significantly expands the mean-variance efficient frontier relative to existing factor models in fifteen out of twenty-five specification tests. When controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDI strategy achieves an alpha of 23 basis points per month with a t-statistic of 2.82.

This paper contributes to the consumption-based asset pricing literature by identifying equity dilution intensity as a novel measure of firms' exposure to aggregate consumption risk, extending the work of Lettau and Ludvigson (2001) on time-varying risk premia and Parker and Julliard (2005) on ultimate consumption risk. Unlike Daniel and Titman (2006) who emphasize behavioral explanations for financing-related anomalies, we demonstrate that dilution intensity captures rational compensation for bearing systematic consumption risk during economic downturns. Our findings complement Li and Zhang (2010) by showing that external financing frictions create cross-sectional variation in required returns through their interaction with the stochastic discount factor rather than through investment-based channels alone. We extend the disaster risk framework of Gabaix (2012) by documenting that firms' reliance on external equity financing creates heterogeneous exposure to rare consumption disasters, generating predictable return variation that existing consumption-based models cannot explain. Methodologically, we employ the rigorous testing protocol of Novy-Marx and Velikov (2023) to demonstrate that EDI's predictive power survives multiple robustness tests including transaction costs, alternative portfolio constructions, and controls for 210 existing anomalies. Our results imply that incorporating firms' financing constraints into consumption-based models can substantially improve their ability to explain cross-sectional return variation, with important implications for cost of capital estimation and portfolio allocation decisions.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their distinct regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine the equity dilution intensity signal across various market conditions and business cycles.construction of our signal relies on two key accounting variables from COMPUSTAT. Stockholders' equity (SEQ) represents the book value of common equity, calculated as total assets minus total liabilities and preferred stock. This measure captures the residual claim of common shareholders on the firm's assets after all obligations have been satisfied. Cash and short-term investments (CHE) represents the sum of cash and all securities readily convertible into cash, including marketable securities and other cash equivalents reported on the balance sheet. This variable measures the firm's most liquid assets available for immediate deployment.construct the Equity Dilution Intensity signal by calculating the year-over-year change in stockholders' equity scaled by lagged cash and short-term investments: (SEQ(t) - SEQ(t-1)) / CHE(t-1). This ratio captures the magnitude of equity base expansion relative to the firm's liquid resources in the prior period. A higher value indicates more intensive equity dilution relative to available cash reserves, potentially signaling either aggressive equity financing activities or rapid internal equity growth through retained earnings. We compute this measure using fiscal year-end values to ensure consistency across firms with different fiscal year endings. To maintain data quality, we require non-missing values for both SEQ and CHE, and exclude observations where CHE(t-1) equals zero to avoid undefined ratios. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDI signal. Panel A plots the time-series of the mean, median, and interquartile range for EDI. On average, the cross-sectional mean (median) EDI is -4.62 (-0.36) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDI data. The signal's interquartile range spans -2.56 to 0.65. Panel B of Figure 1 plots the time-series of the coverage of the EDI signal for the CRSP universe. On average, the EDI signal is available for 6.27% of CRSP names, which on average make up 7.59% of total market capitalization.

4 Does EDI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDI portfolio and sells the low EDI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDI strategy earns an average return of 0.28% per month with a t-statistic of 2.87. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.12% to 0.28% per month and have t-statistics exceeding 1.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 10.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least $1,291 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-40bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.86. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the EDI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDI, as well as average returns and alphas for long/short trading EDI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDI strategy achieves an average return of 33 bps/month with a t-statistic of 3.63. Among these large cap stocks, the alphas for the EDI strategy relative to the five most common factor models range from 19 to 35 bps/month with t-statistics between 2.23 and 3.82.

5 How does EDI perform relative to the zoo?

Figure 2 puts the performance of EDI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDI strategy falls in the distribution. The EDI strategy's gross (net) Sharpe ratio of 0.37 (0.31) is greater than 78% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EDI strategy (red line).[2] Ignoring trading costs, a $1 invested in the EDI strategy would have yielded $NaN which ranks the EDI strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EDI strategy would have yielded $NaN which ranks the EDI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDI relative to those. Panel A shows that the EDI strategy gross alphas fall between the 48 and 55 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDI ranks between the 67 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDI or at least to weaken the power EDI has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDI signal in these Fama-MacBeth regressions exceed 2.20, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on EDI is 1.15.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDI strategy earns alphas that range from 8-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 0.87, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.82.

7 Does EDI add relative to the whole zoo?

Finally, we can ask how much adding EDI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EDI grows to $2761.63.

8 Conclusion

This study provides a comprehensive analysis of Equity Dilution Intensity (EDI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that EDI demonstrates moderate predictive power for equity returns, with a value-weighted

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDI is available.

long/short strategy achieving a gross Sharpe ratio of 0.37 and a net Sharpe ratio of 0.31 after accounting for transaction costs. While the signal shows positive abnormal returns relative to the Fama-French five-factor model plus momentum (12 bps/month gross, 11 bps/month net), the statistical significance remains marginal with t-statistics of 1.31 and 1.19, respectively.

The most compelling evidence for EDI's unique contribution emerges when controlling for closely related factors from the asset pricing literature. After adjusting for six established factors plus six conceptually similar strategies—including change in equity to assets, growth in book equity, asset growth, change in net operating assets, total accruals, and percent total accruals—EDI generates a statistically significant alpha of 23 bps/month (t-statistic = 2.82). This suggests that EDI captures distinct information about future returns beyond what is explained by existing dilution and growth-related anomalies.

From a practical perspective, while EDI's standalone performance may not justify its implementation as an independent trading strategy given the modest Sharpe ratios and marginal statistical significance, its orthogonality to existing factors suggests potential value as a complementary signal in multi-factor portfolios. The persistence of alpha after controlling for related strategies indicates that EDI may capture unique aspects of firms' financing decisions and their implications for future performance that are not fully reflected in traditional growth or accrual measures.

Several limitations warrant consideration in interpreting these results. First, the analysis is confined to the specific sample period and market examined, and the robustness of EDI across different market regimes and international markets remains unexplored. Second, while we control for transaction costs, implementation challenges such as market impact and capacity constraints for institutional investors require further investigation. Third, the economic mechanisms underlying EDI's predictive power deserve deeper theoretical and empirical examination.

Future research should pursue several promising directions. Investigating the time-varying nature of EDI's predictive power could reveal whether its effectiveness depends on market conditions, monetary policy regimes, or investor sentiment. Cross-sectional heterogeneity analysis could identify firm characteristics that moderate the EDI-return relationship, potentially enhancing the signal's efficacy. Additionally, examining the interaction between EDI and corporate events such as mergers, acquisitions, or strategic shifts could provide insights into the economic channels through which equity dilution affects future returns. Finally, extending the analysis to international markets would test the generalizability of our findings and potentially uncover market-specific factors that influence the relationship between equity dilution and returns.

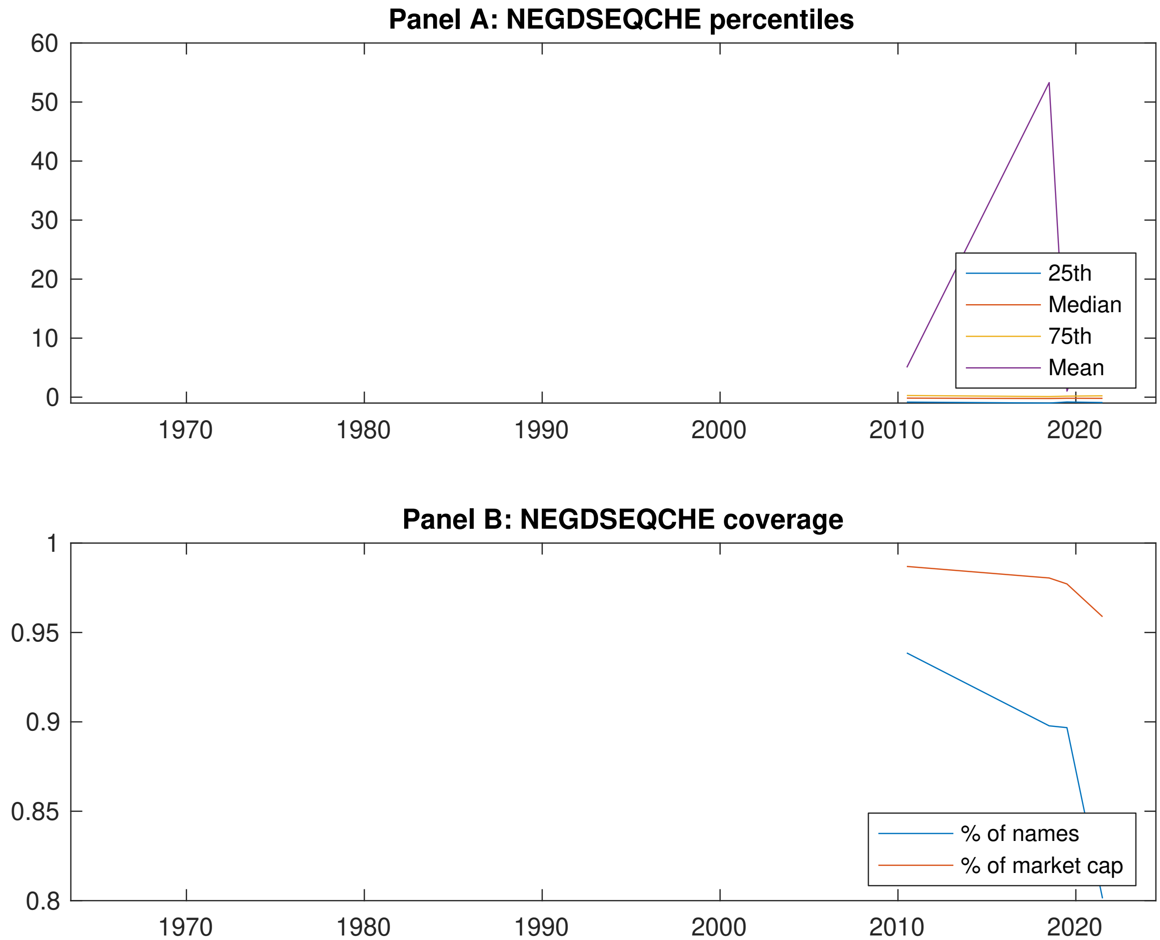


Figure 1: Times series of EDI percentiles and coverage.
This figure plots descriptive statistics for EDI. Panel A shows cross-sectional percentiles of EDI over the sample. Panel B plots the monthly coverage of EDI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.38	0.61	0.66	0.64	0.67	0.28
	[2.21]	[3.47]	[3.88]	[3.72]	[3.81]	[2.87]
α_{CAPM}	-0.19	0.02	0.09	0.05	0.10	0.28
	[-3.35]	[0.30]	[2.18]	[1.06]	[1.36]	[2.90]
α_{FF3}	-0.21	0.05	0.14	-0.01	0.01	0.21
	[-3.60]	[0.86]	[3.37]	[-0.16]	[0.20]	[2.20]
α_{FF4}	-0.20	0.06	0.12	0.01	-0.00	0.19
	[-3.43]	[1.10]	[2.94]	[0.16]	[-0.01]	[1.94]
α_{FF5}	-0.23	0.06	0.15	-0.05	-0.09	0.12
	[-3.99]	[1.09]	[3.57]	[-1.00]	[-1.33]	[1.36]
α_{FF6}	-0.23	0.07	0.13	-0.03	-0.09	0.12
	[-3.86]	[1.23]	[3.17]	[-0.63]	[-1.30]	[1.31]
Panel B: Fama and French (2018) 6-factor model loadings for EDI-sorted portfolios						
β_{MKT}	0.97	0.99	0.99	1.05	1.03	0.06
	[68.69]	[73.34]	[98.02]	[90.79]	[61.93]	[2.86]
β_{SMB}	0.06	-0.04	-0.09	-0.06	0.11	0.05
	[2.77]	[-2.08]	[-6.29]	[-3.44]	[4.57]	[1.71]
β_{HML}	0.10	-0.01	-0.05	0.12	0.00	-0.10
	[3.57]	[-0.33]	[-2.74]	[5.53]	[0.06]	[-2.28]
β_{RMW}	0.15	0.05	0.00	0.05	0.05	-0.09
	[5.27]	[1.99]	[0.08]	[2.35]	[1.53]	[-2.14]
β_{CMA}	-0.15	-0.16	-0.08	0.13	0.48	0.63
	[-3.70]	[-4.16]	[-2.89]	[3.87]	[10.14]	[10.02]
β_{UMD}	-0.01	-0.01	0.02	-0.02	-0.00	0.00
	[-0.43]	[-0.94]	[2.23]	[-2.14]	[-0.09]	[0.19]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	590	673	765	768	
me ($\$10^6$)	1366	2129	2782	2606	1291	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EDI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.28	0.21	0.19	0.12	0.12
			[2.87]	[2.90]	[2.20]	[1.94]	[1.36]	[1.31]
Quintile	NYSE	EW	0.48	0.48	0.40	0.42	0.48	0.50
			[4.28]	[4.20]	[3.73]	[3.82]	[4.70]	[4.81]
Quintile	Name	VW	0.25	0.26	0.18	0.16	0.08	0.09
			[2.58]	[2.66]	[1.84]	[1.62]	[0.93]	[0.91]
Quintile	Cap	VW	0.25	0.26	0.21	0.21	0.13	0.15
			[3.34]	[3.59]	[2.93]	[2.89]	[2.02]	[2.23]
Decile	NYSE	VW	0.46	0.48	0.37	0.33	0.23	0.22
			[4.25]	[4.45]	[3.61]	[3.13]	[2.43]	[2.23]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.24	0.24	0.17	0.16	0.11	0.11
			[2.45]	[2.45]	[1.81]	[1.68]	[1.23]	[1.19]
Quintile	NYSE	EW	0.27	0.26	0.19	0.21	0.24	0.25
			[2.30]	[2.23]	[1.71]	[1.84]	[2.28]	[2.42]
Quintile	Name	VW	0.21	0.23	0.15	0.14	0.08	0.08
			[2.15]	[2.28]	[1.55]	[1.44]	[0.90]	[0.89]
Quintile	Cap	VW	0.21	0.24	0.19	0.19	0.13	0.14
			[2.86]	[3.20]	[2.60]	[2.61]	[2.04]	[2.17]
Decile	NYSE	VW	0.40	0.43	0.33	0.31	0.23	0.22
			[3.74]	[3.96]	[3.21]	[2.96]	[2.43]	[2.31]

Table 3: Conditional sort on size and EDI

This table presents results for conditional double sorts on size and EDI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDI and short stocks with low EDI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDI Quintiles					EDI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.51	0.80	0.95	0.92	0.77	0.25	0.20	0.11	0.08	0.17	0.15
		[2.07]	[3.35]	[4.22]	[3.46]	[2.57]	[1.71]	[1.34]	[0.79]	[0.58]	[1.25]	[1.10]
	(2)	0.62	0.72	0.85	0.93	0.90	0.26	0.28	0.21	0.21	0.31	0.31
		[2.59]	[3.12]	[3.77]	[4.30]	[3.72]	[2.46]	[2.62]	[2.03]	[1.97]	[3.29]	[3.23]
	(3)	0.61	0.78	0.77	0.90	0.85	0.24	0.24	0.17	0.14	0.25	0.22
		[2.80]	[3.60]	[3.69]	[4.43]	[3.88]	[2.21]	[2.28]	[1.65]	[1.34]	[2.51]	[2.24]
	(4)	0.49	0.68	0.79	0.88	0.82	0.34	0.32	0.22	0.26	0.16	0.20
		[2.47]	[3.43]	[3.89]	[4.55]	[4.09]	[3.72]	[3.51]	[2.56]	[2.98]	[1.85]	[2.37]
	(5)	0.34	0.59	0.61	0.54	0.66	0.33	0.35	0.30	0.30	0.19	0.21
		[1.99]	[3.57]	[3.55]	[2.99]	[4.11]	[3.63]	[3.82]	[3.26]	[3.24]	[2.23]	[2.42]

Panel B: Portfolio average number of firms and market capitalization

		EDI Quintiles					EDI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	384	385	385	379	358	37	39	34	30	24
	(2)	107	108	108	107	105	57	58	57	57	55
	(3)	78	78	78	78	78	100	99	101	100	99
	(4)	65	65	65	65	65	213	211	218	213	214
	(5)	60	59	60	60	59	1201	1696	1936	1888	1508

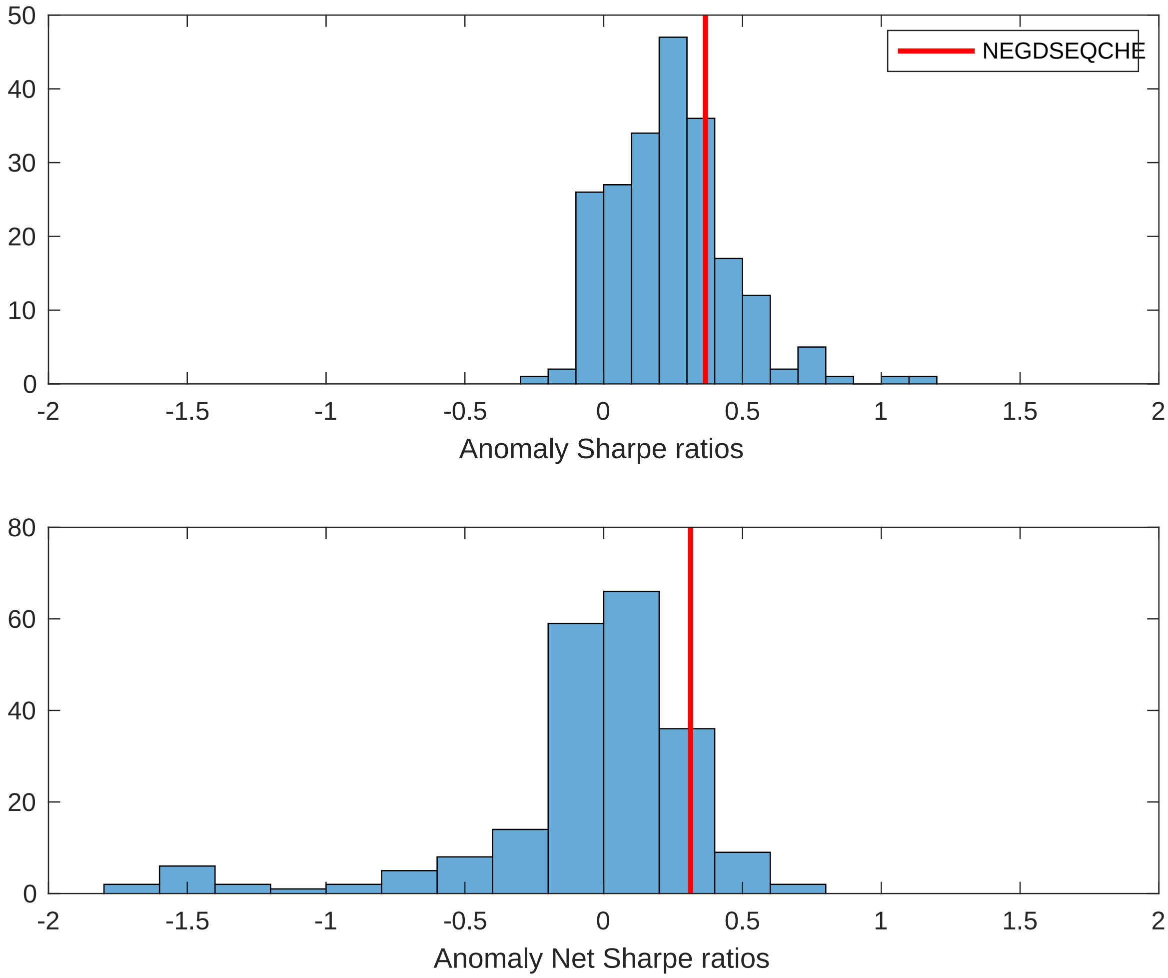


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

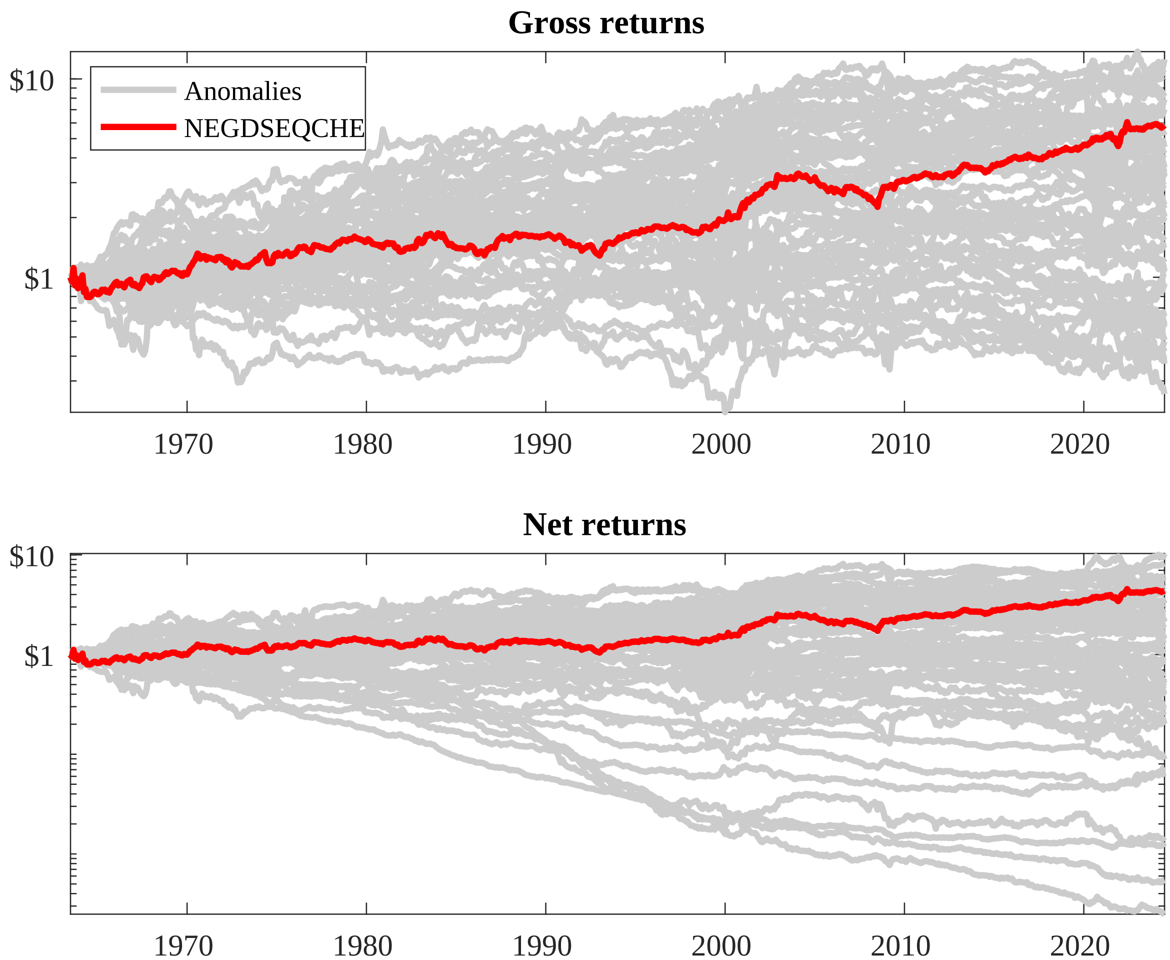
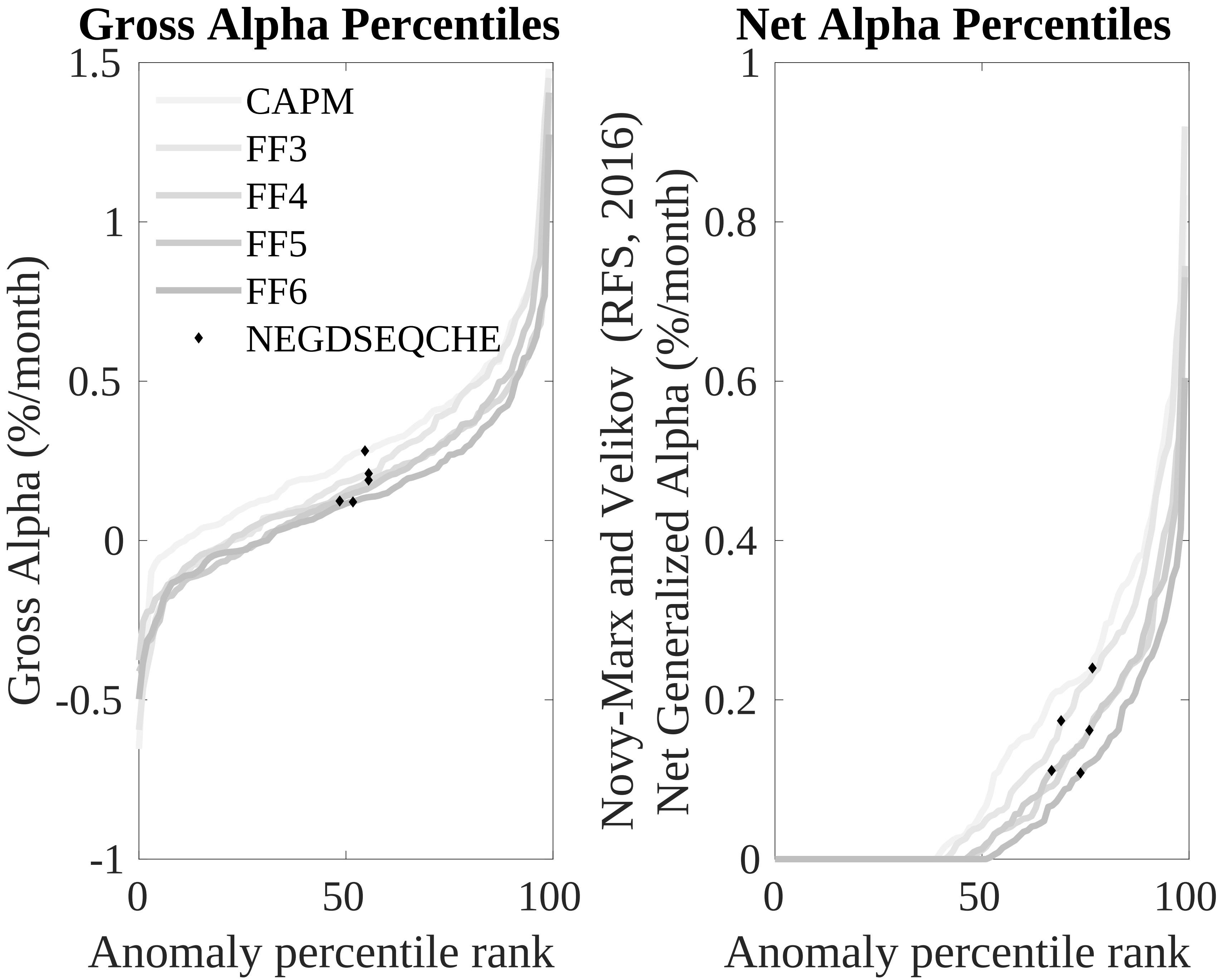


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDSEQCHE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

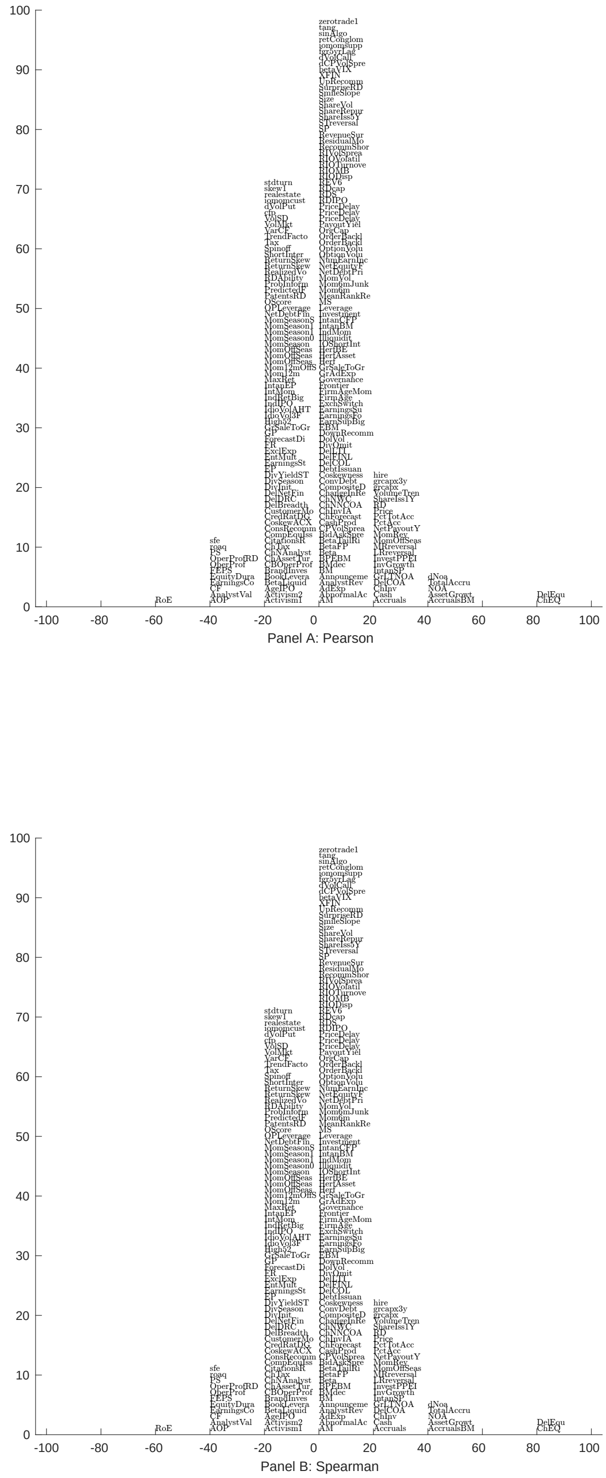


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

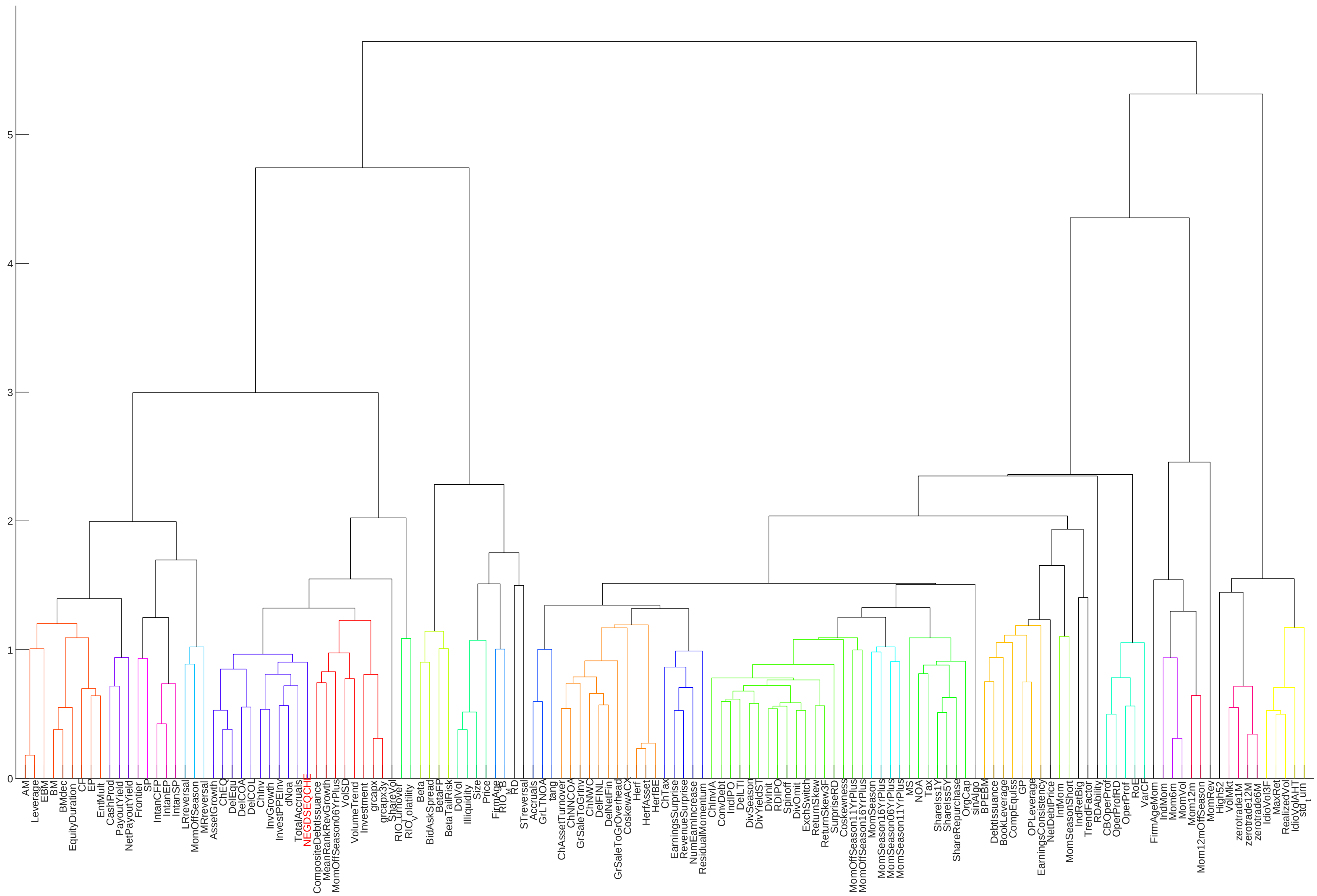


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

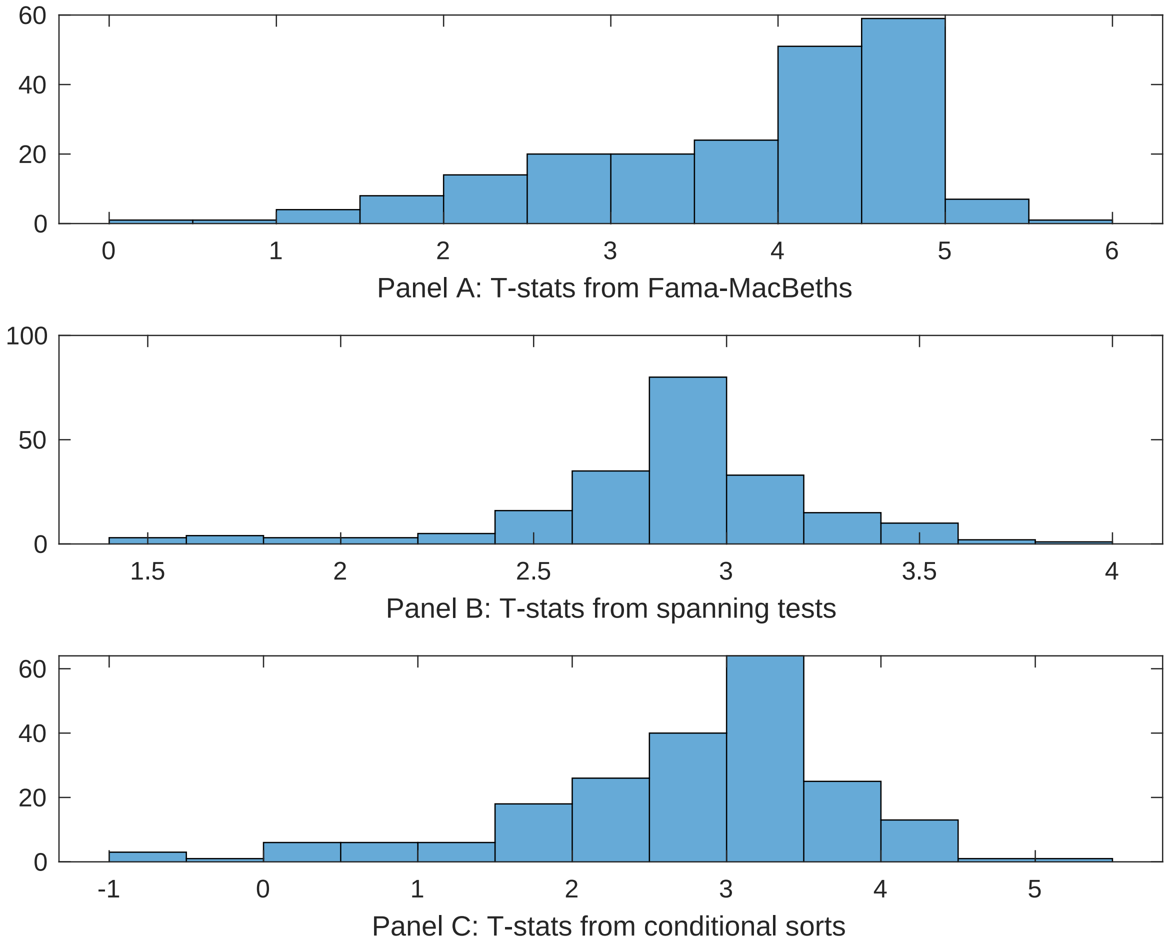


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.17	0.13	0.13	0.12	0.11	0.12
	[5.67]	[7.39]	[6.10]	[5.94]	[5.40]	[3.92]	[5.05]
EDI	0.14	0.16	0.10	0.13	0.24	0.12	0.36
	[3.15]	[3.60]	[2.20]	[2.52]	[4.43]	[3.34]	[1.15]
Anomaly 1	0.14						0.62
	[4.00]						[1.50]
Anomaly 2		0.44					0.12
		[4.31]					[0.13]
Anomaly 3			0.99				0.48
			[9.29]				[3.73]
Anomaly 4				0.13			0.41
				[9.91]			[2.36]
Anomaly 5					0.34		-0.25
					[1.82]		[-1.02]
Anomaly 6						0.13	0.59
						[3.11]	[1.37]
# months	708	708	708	708	708	432	432
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.10	0.12	0.08	0.09	0.26	0.23
	[1.48]	[1.15]	[1.36]	[0.87]	[1.03]	[2.87]	[2.82]
Anomaly 1	41.89						10.17
	[9.60]						[1.24]
Anomaly 2		47.92					47.47
		[10.11]					[6.50]
Anomaly 3			23.29				-21.05
			[4.24]				[-3.59]
Anomaly 4				29.57			13.79
				[5.77]			[2.67]
Anomaly 5					20.50		-1.40
					[4.85]		[-0.27]
Anomaly 6						19.94	11.17
						[4.95]	[2.70]
mkt	5.67	8.01	6.47	6.58	5.73	-0.55	1.25
	[2.70]	[3.84]	[2.94]	[3.02]	[2.61]	[-0.24]	[0.61]
smb	5.49	4.42	4.16	6.34	6.11	4.84	2.99
	[1.81]	[1.46]	[1.30]	[2.02]	[1.93]	[1.48]	[1.01]
hml	-14.74	-15.02	-9.91	-11.46	-8.96	-5.97	-9.73
	[-3.60]	[-3.69]	[-2.33]	[-2.71]	[-2.12]	[-1.49]	[-2.61]
rmw	-5.61	-7.19	-9.78	-9.20	-6.21	-13.58	-17.18
	[-1.37]	[-1.77]	[-2.28]	[-2.17]	[-1.44]	[-3.25]	[-4.52]
cma	20.34	16.08	34.42	40.47	52.23	51.70	15.36
	[2.75]	[2.15]	[3.78]	[5.57]	[7.95]	[8.67]	[1.99]
umd	1.46	-0.38	1.21	-0.13	0.98	0.61	-1.85
	[0.70]	[-0.18]	[0.55]	[-0.06]	[0.45]	[0.30]	[-0.99]
# months	731	731	731	731	731	438	438
$\bar{R}^2(\%)$	27	28	19	21	20	31	46

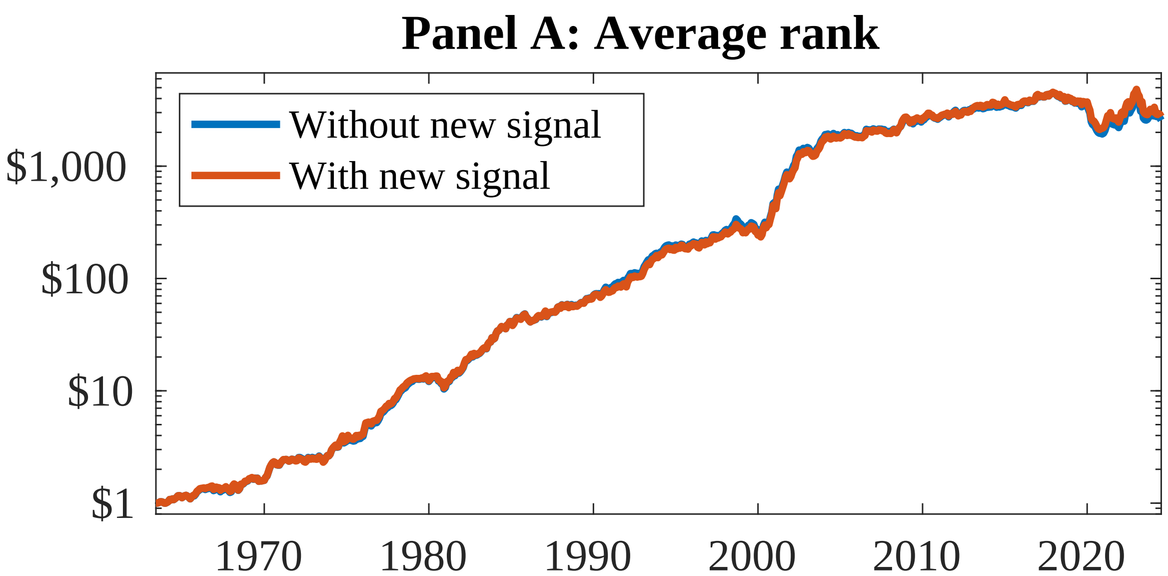


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. (2005). *Asset Pricing: Revised Edition*. Princeton University Press.

Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming*.

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.

Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.

Li, E. X. and Zhang, L. (2010). A neoclassical interpretation of momentum. *Journal of Financial Economics*, 97(3):472–490.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.

Parker, J. A. and Julliard, C. (2005). Consumption risk and the cross section of expected returns. *Journal of Political Economy*, 113(1):185–222.

Wachter, J. A. (2006). A consumption-based model of the term structure of interest rates. *Journal of Financial Economics*, 79(2):365–399.